Statistical Data Value Chain

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Value Chains Introduction

Purpose of this chapter

This chapter describes how the statistical value chain works in practice. The Strategy chapter introduced the ten value streams, their stages, and the value they create. The Steady-State Architecture describes the results the chain hands over: the states, their quality gates, and their metadata. This chapter describes the work between those results: the stages, the processes that realise them, the roles that carry them out, and the events that connect them. Every value stream is described with the same structure, so that NSIs can compare them and map their own organisation onto them. **[VC1]** ### Where to find what

Question	Answered in
Which value streams exist, and what value do they create?	Strategy, *Value Streams*
What must data satisfy at each handover point?	Steady-State Architecture, per state
How is the work in each value stream organised?	This chapter
Which actors, roles, processes, and functions exist?	Business Architecture, reference content
How mature is an NSI, and how does it apply a value stream?	SAF Toolkit

The division of work between states and value streams is deliberate. A state describes the **result**: what the data looks like, which gate it passed, and how it is protected. A value stream describes the **work**: who does what, in which order, and with which controls. Gate criteria and minimum metadata sets are therefore not repeated in this chapter. Each value stream refers to the state it produces.

Key concepts — if you are new to architecture documents This chapter uses a handful of architecture terms. Here is what they mean in plain language; the glossary has the full definitions.

- **Value stream** — a sequence of stages that together create a statistical output. Think of it as a production line: VS-01 to VS-06 each add a specific kind of value to the data.
- **Value stream stage vs. business process** — the stage (`VS-0x.y`) says *what value* is added; the business processes (`PRD`, `DLM`, …) describe *how* the work is done. Attribute 6 of each value stream shows which processes realise each stage.
- **Steady state** — a fixed handover point: a versioned, quality-assured dataset that one stream produces and the next one consumes. There are five numbered states, #1 to #5; data still at the external provider (#0) and data in users' hands (#6) are boundary positions outside the NSI's control.
- **Business service / object / event** — what a value stream offers to others / the data artefacts it works on / the signals streams send each other (e.g. *Raw Data Registered* tells the next stream new data is ready).
- **GSBPM, GAMSO** — the international models of statistical production and of the activities of statistical organisations (see *Relationship with statistical standards* in the Introduction).

The chain at a glance

Six primary value streams form the production chain. Each one consumes a registered state version and produces the next. They are loosely coupled (SAF-P07): a value stream needs no knowledge of the internal work of its neighbors, only of the state it receives and the state it delivers. The handover is triggered by a registration event. Four supporting value streams run continuously across the whole chain. They set the rules, develop new methods, provide the platforms, and keep the chain connected to its users and data providers.

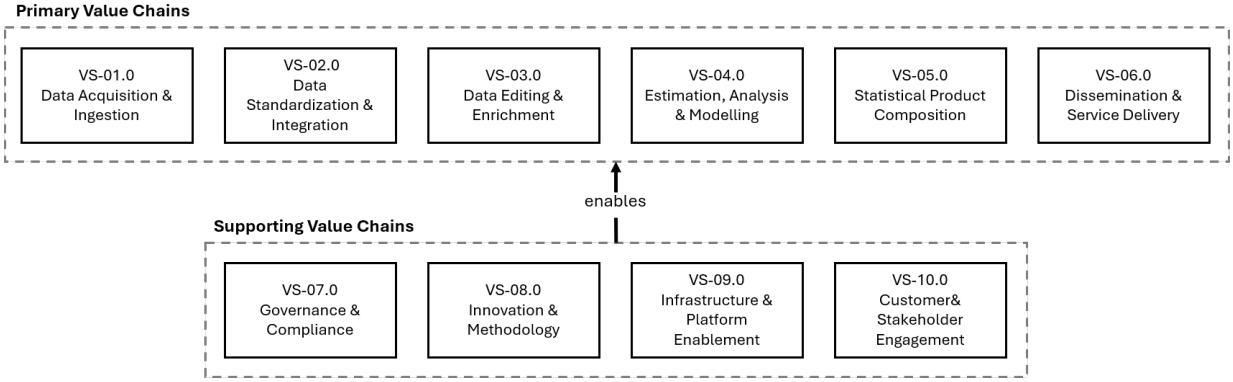


Figure 1: Value Stream Context Map

VS	Value stream	Type	State in → out	Event out	Signature discipline or contribution
VS-01	Data Acquisition and Ingestion	Primary	#0 → #1	*Raw Data Registered*	Metadata and validation at ingestion
VS-02	Data Standard-ization and Integration	Primary	#1 → #2	*Standardized Data Registered*	No loss of content: map, reformat, annotate
VS-03	Data Editing and Enrichment	Primary	#2 → #3	*Processed Data Registered*	Full traceability of every value change

VS	Value stream	Type	State in → out	Event out	Signature discipline or contribution
VS-04	Estimation, Analysis and Statistical Modeling	Primary	#3 → #4 (analytical)	*Output Data Registered*	Methodological transparency, accuracy, and disclosure control on outputs
VS-05	Statistical Product Composition	Primary	#4 (analytical) → #4 (approved)	*Statistical Product Registered*	Coherence and interpretability; approval before registration
VS-06	Dissemination and Service Delivery	Primary	#4 (approved) → #5	*Statistics Released*	Accessibility and trust; final disclosure-control gate
VS-07	Governance and Compliance	Supporting	Across the chain	n/a	Frameworks, controls, and assurance; enforces the data principles
VS-08	Innovation and Methodological Development	Supporting	Across the chain	n/a	Validated methods and tools transferred into production
VS-09	Infrastructure and Platform Enablement	Supporting	Across the chain	n/a	Platforms, environments, integration, and continuity
VS-10	Customer and Stakeholder Engagement	Supporting	Across the chain; feedback from #5 and #6	n/a	Understood needs, service portfolio, relationships, and impact

Not every statistical product passes through every primary value stream. A macro statistic, for example, often starts from statistics another domain has already registered. Such a product enters the chain downstream and its lineage points to the state it consumed (see *Applying the steady states pragmatically*). Data that comes from outside the NSI always enters through VS-01.

How the supporting value streams differ

The four supporting value streams are not stages of the production line. They work across the whole chain and enable and constrain all six primary value streams continuously. They differ from the primary value streams in four ways:

- **No state transition.** A supporting value stream does not take data from one state to the next, so it has no registration event and no boundary interfaces of its own. It interacts with the whole chain at once.
- **A continuous cycle.** It runs as a cycle of plan, build, operate, and improve, not per production cycle.
- **An enabling output.** Its output is a capability, not data: governance and assurance (VS-07), validated methods (VS-08), platforms (VS-09), and understood needs and relationships (VS-10).
- **GAMSO instead of GSBPM.** It aligns with the GAMSO activity areas Strategy and Leadership, Capability Management, and Corporate Support, and it foregrounds principles SAF-P02 to SAF-P06.

The diagram shows both directions of the interaction: the supporting value streams provide capabilities to the chain, and the chain sends back evidence, needs, and usage.

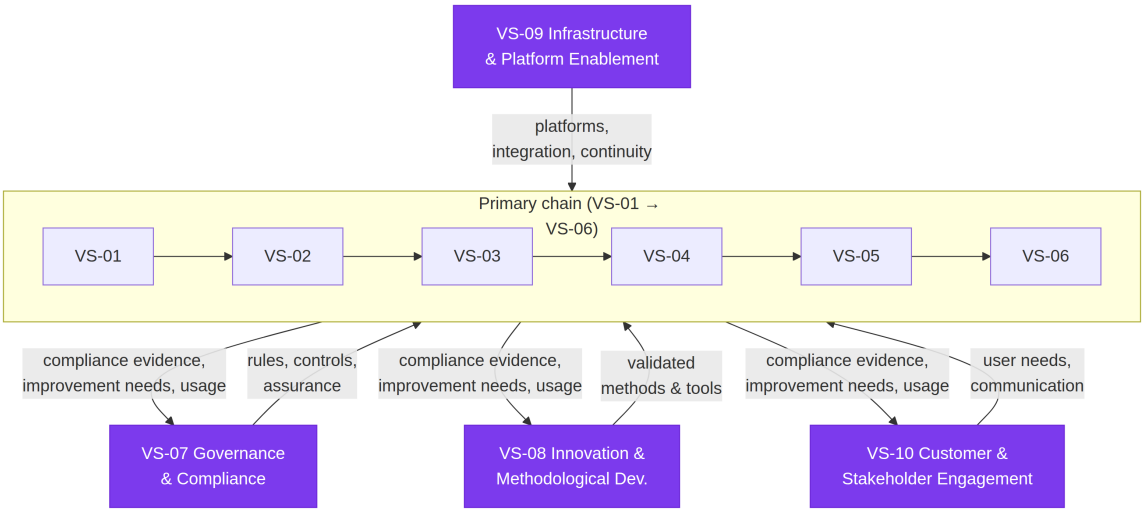


Figure 2: The four supporting streams (purple) and the primary chain: each provides a capability to the chain and receives feedback or needs from it.

The four process types

Every business domain, and therefore every primary value stream, runs the same four process types. **Design** decides how the work is done and delivers metadata: designs, rules, and specifications. **Implementation** builds and tests the solution and delivers the execution process. **Execution** runs the work in each production cycle and delivers the statistical data; the stages of a value stream describe this execution. **Management** monitors the work, assesses quality, and feeds improvements back to design. Design and implementation only run when something changes, execution runs every cycle, and management runs continuously.

The supporting value streams do not follow a production cycle. They run the same pattern as a continuous cycle of plan, build, operate, and improve.

Process type	Provides	GSBPM phases	Equivalent in supporting streams
Design	Metadata: designs, rules, specifications	1 Specify Needs, 2 Design	Plan
Implementation	The execution process: instruments, components, configured workflows	3 Build	Build
Execution	Statistical data, registered in a state	4 Collect, 5 Process, 6 Analyse, 7 Disseminate	Operate
Management	Quality reports and improvements	8 Evaluate; overarching quality and metadata management	Improve

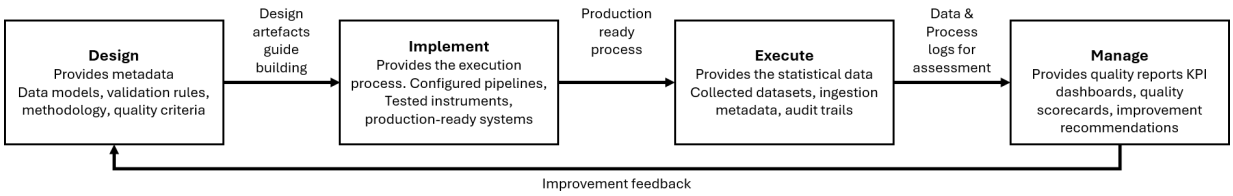


Figure 3: The Four-Process Governance Pattern

How each value stream is described

Each primary value stream is described with the same ten attributes. The supporting value streams use the same attributes, with two differences: attribute 2 describes the relation to the chain instead of a state transition, and attribute 4 follows the cycle of plan, build, operate, and improve.

#	Attribute	Question it answers
1	Purpose and value created	Why does this value stream exist, and what value does it add?
2	Position and state transition	Which state does it consume, and which does it produce?
3	Stages	Which stages create the value, and what happens in each?
4	The four process types	How is the value stream designed, built, and managed?
5	Actors and roles	Who is involved, and with which responsibility?
6	Capabilities, processes, and functions	Which SAF capabilities, processes, and functions realise the stages?
7	Principles	Which architecture principles weigh most, and how are they applied?
8	Interfaces, services, and events	What does the value stream offer and exchange, and what triggers it?
9	Key performance indicators	How is the performance of the value stream measured?
10	Examples	What does the value stream look like in practice?

Each value stream has three ArchiMate views: (a) the value stream with its stages and the capabilities that serve them, (b) the stages and the processes and functions that realise them, and (c) the state handover and its events. Maturity checklists and detailed end-to-end scenarios for each value stream are part of the SAF Toolkit. **[VC2]**

Readers can go straight to the attributes that concern them. Operations staff read the stages in attribute 3. Methodologists and statistical designers read the Design processes, system builders the Implementation processes, and quality and compliance managers the Management processes, all in attribute 4. Managers who want the big picture read attributes 1, 2, and 10.

Standard activities in every value stream

Some sub-processes work the same way in every primary value stream. They are described once here. Each value stream lists them in its process tables, marked *(standard)*, with only its own outputs and additions.

Test and Validate the Pipeline (Implementation). Before a process goes into production, it is verified in full. Technical testing covers each component, and integration testing checks the end-to-end flow from

the input state to registration of the output state. A pilot with real or synthetic data verifies that metadata is captured and propagated, that the quality gate is enforced, and that exceptions are handled as designed. Performance testing confirms that expected volumes fit the agreed time windows. Security testing makes sure the process meets SAF-P10. The sub-process ends with putting the production system into use and documenting the operational procedures.

Activity	Key outputs	SAF process	Other reference
Test all individual components	Unit test reports	PRD-3.5 Test production systems	
Test integration: end-to-end flow from input state to registration	Integration test report	PRD-3.5 Test production systems	
Run a pilot with real or synthetic data	Pilot report, validated metadata propagation	PRD-3.6 Test statistical business process	
Test performance against expected volumes	Performance test report	PRD-3.5 Test production systems	
Test security: penetration testing, access control verification	Security test report	PRD-3.5 Test production systems	SAF-P10; 3.3.2 Security planning
Finalise the production system and document operational procedures	Operational procedures manual, production-ready process	PRD-3.7 Finalise production systems	

Monitor Process Performance (Management). The process owner and the quality manager collect the key performance indicators of the value stream (attribute 9) in every cycle. They track paradata: who performed each step, when, how long it took, which tools were used, and which errors occurred. Dashboards give the process owner a current view of the health of the value stream. The approach follows the Quality Management activity of GAMSO.

Activity	Key outputs	SAF process	Other reference
Collect and aggregate the key performance indicators per cycle	Process performance dashboard	OVA-4 Process Data Management; PRD-8.1 Gather evaluation inputs	GAMSO Quality Management
Track paradata: operator, timing, tools, errors	Paradata records	OVA-4 Process Data Management	
Produce process performance dashboards	Dashboard for the process owner	OVA-4 Process Data Management	

Audit and Compliance Reporting (Management). This sub-process produces the evidence that the value stream operates within legal, regulatory, and organisational requirements: audit trail reports that trace every artefact from its input state to its output state, data protection reporting, and security compliance reporting, including breaches and incidents. The reports meet the obligation of capability 4.1.5 *Accountability* and feed VS-07.5 *Audit, Certify and Report Compliance*.

Activity	Key outputs	SAF process	Other reference
Produce audit trail reports: full traceability from input to output state	Audit trail summary	OVA-4 Process Data Management; DLM-8 Monitor data quality, lineage and use	4.1.5 Accountability

Activity	Key outputs	SAF process	Other reference
Report on data protection compliance	Data protection compliance report	OVA-1 Quality Management	SAF-P11; 3.3.1 Privacy management
Report on security compliance: measures applied, breaches, incidents	Security compliance report (per cycle)	OVA-1 Quality Management	SAF-P10; 3.3.3 Security incident management

Continuous Improvement (Management). This sub-process closes the governance cycle. It analyses quality and performance trends across cycles, identifies improvement opportunities, prioritises them by impact and feasibility, and feeds them back to Design. Where a new method, source, or tool is needed, the recommendation goes to VS-08. This is capability 4.1.4 *Improvement management* in practice, and connects to the GAMSO idea of transferring mature results of innovation to production.

Activity	Key outputs	SAF process	Other reference
Analyse quality and performance trends across cycles	Improvement recommendations report	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Identify improvement opportunities	Updated design inputs	PRD-8.2 Conduct evaluation	4.1.4 Improvement management
Prioritise improvements by impact and feasibility	Improvement tracking log	PRD-8.3 Agree an action plan	GAMSO Capability Management
Feed recommendations back to Design or to VS-08, and track their implementation	Design feedback (closing the cycle)	PRD-8.3 Agree an action plan	4.1.4 Improvement management

Paradata. Every process type records how it ran, in addition to the data metadata listed per value stream.

Process type	Process metadata (paradata) produced
Design	Log of design decisions, methodology version, review and approval records
Implementation	Build log, test results, records of configuration changes, deployment approvals
Execution	Operator identity, timing, volumes, errors, tool and rule versions applied
Management	Dashboards of key performance indicators, quality trend analyses, improvement decisions, audit findings

Boundary interfaces. Each primary value stream is loosely coupled (SAF-P07) to its neighbors through boundary interfaces. Each interface exchanges a registered, immutable artefact and its metadata, never internal working data, so neither side needs knowledge of the other's internal processing. Attribute 8 of each value stream specifies its boundaries: the artefact handed over, the metadata it must carry, the triggering event, and the point where responsibility transfers.

Identifiers used in this chapter

Identifier	Refers to	Defined in
VS-01 to VS-10, VS-01.1, …	Value streams and their stages	Strategy, *Value Streams*
1.2.1, 2.1.2, …	Capabilities	Strategy, *Capability Model*

Identifier	Refers to	Defined in
ACT, ROL, COL	Business actors, roles, and collaborations	Business Architecture
PRD, OVA, DLM, MDS, DSP	Business processes	Business Architecture, *Business Processes*
FUN	Business functions	Business Architecture, *Business Functions*
#0 to #6	Steady states and boundary positions	Steady-State Architecture
SAF-P01 to SAF-P11	Architecture principles	Motivation, *Architecture Principles*

Steady States of Data

The states that the value streams consume and produce are defined in the Steady-State Architecture: their quality gates, minimum metadata, the rules common to all states, and when a state may be shared or skipped (see Applying the steady states pragmatically). This chapter does not repeat them.

Principles to Apply

The SAF defines eleven architecture principles, labeled SAF-P01 to SAF-P11. The six **primary** streams actively apply six of them (marked below); the four **supporting** streams foreground the other five (marked †). The table shows practical guidance on how to apply each principle; the statements, rationale and implications are in *Architecture Principles* (Motivation); VS-07 in particular **enforces** the data principles (P09/P10/P11) across the whole chain.

Table 1: SAF Principles

Principle	Name	How to apply across the value chain
SAF-P01	Open-source participation	Build statistical capabilities on shared, open tooling; contribute back, so NSIs keep collective control of critical methods rather than depending on vendors.
SAF-P07	Loose coupling	Exchange data only at steady-state interfaces; each stream consumes one state and produces the next without knowing the others' internals.
SAF-P08	Data is shared property	Once an artefact is registered at a steady state, make it available to all authorised consumers; avoid duplicating collection, processing, or analysis.
SAF-P09	No data without metadata and classification	Every artefact carries provenance (origin), methodology, accuracy, classification, and lineage (the trace back to its inputs) — captured at the point each value is created or changed.
SAF-P10	Security by design	Embed access control, integrity verification, and tamper-evident audit trails in every pipeline from the start.
SAF-P11	Privacy by design	Remove/pseudonymise identifiers during standardisation; assess disclosure risk on enrichment and apply statistical disclosure control before release.
SAF-P02 †	Active support	Operate and support capabilities as living services, not one-off builds (VS-09 platforms, VS-10 services).
SAF-P03 †	Active life cycle management	Version and govern rules, methods, models, and products across their life cycle; retire what is obsolete (VS-07, VS-08).
SAF-P04 †	Maximise the benefits for the organisation	Prefer solutions that serve the whole organisation and real user value over local optimisations (VS-10, VS-07).
SAF-P05 †	IT and technology is everyone's business	Treat methodology, data, and technology as a shared responsibility across business and IT (VS-08, VS-09).

Principle	Name	How to apply across the value chain
SAF-P06 †	Business continuity	Keep the chain running reliably and able to recover from disruption (VS-09).

= actively applied by the six primary value streams. † = foregrounded by the four supporting value streams (VS-07–VS-10).

Metadata everywhere (SAF-P09): value is only added once its context is captured — what was done, by which rule/method, with what accuracy, and traceable to its inputs.

Confidentiality progressively applied (SAF-P11): identifiers are removed/pseudonymised at standardisation (#2) at the latest, or earlier where that suits the process; disclosure risk is assessed on enrichment (#3), disclosure control is applied to outputs (#4), and a final disclosure check gates release (#5).